

# A Self-Hosted Measurement Substrate Resolves a Capability Cliff Into a Two-Parameter Logistic Law

Stepwise scaling on a 0\$-incremental Qwen2.5 ladder (0.5B / 1.5B / 3B / 7B): all four sections (§Formula, §Method, §Benchmark, §Benefit) graduated to •  
SUPPORTED-NUMERICAL

hexa-codex maintainers  
[github.com/dancinlab/hexa-codex](https://github.com/dancinlab/hexa-codex)

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## Abstract

We report a capability-evaluation finding measured on a self-hosted measurement substrate — an 0\$-incremental, deterministic, locally served stack the **hexa-codex** project uses as the shared empirical-contact surface for all of its domains. On a stratified  $N=2000$  reading-comprehension manifest evaluated across a four-rung Qwen2.5 parameter ladder (0.5B, 1.5B, 3B, 7B), the diagnostic `wc_31_60` difficulty stratum exhibits a sharp *stepwise* capability cliff: accuracy stays flat at {6%, 16%, 13%} across the three sub-7B rungs and then jumps to 56% at 7B, a single-rung lift of +43 pp. We show this curve is the wrong shape for a single-exponent power law — the best-per-stratum mean-accuracy log-log slope is only  $\hat{\sigma} = 0.172$ , whose monotone 3B  $\rightarrow$  7B increment is just +2.1 pp, one-twentieth of the measured step — and fit instead the simplest closed form that admits a step: a two-parameter logistic (sigmoid) in  $\log_2(\text{params})$  with both plateaus pinned to the data. The fit ( $k=8.95$ ,  $x_0=1.973$ ) recomputes the measured ladder with RMSE = 0.0356, *below* the mean  $N=30$  binomial sampling-noise band of 0.0665; the within-plateau scatter is therefore noise, not structural misfit. The fitted cliff midpoint  $x_0$  lands strictly inside the 3B–7B parameter gap, matching the empirical crossing. A hexa-native verifier recomputes all five checks from the fitted parameters against the committed bench data (`.verdicts/sandbox/m4_substrate_formula_fit.txt`, 5/5 PASS). Operationally, the substrate’s routing-viability flag inverts `false`  $\rightarrow$  `true`  $\rightarrow$  `false` across the ladder, locating cheap-persona routing arbitrage only in a narrow 1.5B–3B mid-scale window. We close by scoping — but explicitly not claiming — the multimodal capability-evaluation extension that the substrate’s release milestone names as remaining work.

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## 1 Introduction

The shape of the capability-versus-scale curve is the central object of the “emergent abilities” debate in large language models. One camp reports sharp, discontinuous jumps in downstream task accuracy as model scale crosses some threshold [6]; a prominent critique argues that such “emergence” is largely a *measurement artifact* of discontinuous or thresholded metrics, and that under smooth, per-token metrics the underlying capability rises continuously [5]. Both positions are usually argued on large, expensive, non-deterministic, API-gated model families where the measurement surface — the exact decoding configuration, the seed, the scorer’s tokenization — is not under the experimenter’s control.

This paper makes a narrow, fully reproducible contribution to that debate from the opposite end of the cost spectrum. We measure a single capability curve on a *self-hosted measurement substrate*: a locally served open-weights model ladder with seed-pinned greedy decoding, a fixed byte-exact scorer, and a committed input manifest, all at 0\$ incremental cost on a consumer Apple-silicon machine. On this substrate the curve we observe *is* stepwise for one specific difficulty stratum, and we ask the constructive question the debate often skips: *if* a step is present in the measured data, what is the simplest closed-form law that reproduces it, and can that law be recomputed by an independent verifier from committed numbers? Our answer is a two-parameter logistic, recomputed to within the bench’s own binomial sampling noise.

We emphasise three properties that distinguish this from a typical scaling-law plot. First, the substrate is the contribution as much as the finding: every number is produced by a serving process we own, so logits, determinism, and unmetered manifest scale are all available, and re-runs are byte-identical. Second, the paper is gated by a project rule (`cx_paper_gate`) that forbids shipping any section whose claim is not recompute-verified to a numerical (green) or formal (blue) tier; the §Formula law here clears that gate via a hexa-native recompute, not a narrative argument. Third, we are explicit about what the substrate does *not* yet measure — the multimodal capability-evaluation extension named by the release milestone is scoped in §8 as remaining work, not folded into the claims.

### Contributions.

- A measured, 0\$-incremental, deterministic four-rung capability ladder on a stratified  $N=2000$  manifest, exhibiting a +43 pp single-rung stepwise cliff on the `wc_31.60` stratum (§5).
- A two-parameter logistic law for that cliff, with both plateaus pinned to the data, recomputed by an independent hexa-native verifier to within the bench’s binomial sampling

noise (§3); and a structural argument, made quantitative, for why a single-exponent power law is the wrong functional family here.

- A routing-window finding: cheap-persona routing arbitrage exists only in a narrow mid-scale band, collapsing at both low and high capability (§6).
- An honest limitations section and an explicit, separately-scoped multimodal residual (§7, §8).

## 2 Related Work

**Scaling laws.** Kaplan et al. [3] established smooth power-law relationships between cross-entropy loss and model size, dataset size, and compute; Hoffmann et al. [2] (Chinchilla) refined the compute-optimal allocation between parameters and tokens. These laws are stated on *loss*, a smooth per-token quantity, and are monotone and scale-free by construction. Our §3 does not contradict them: we fit a *downstream-accuracy* curve on a thresholded task scorer, which need not inherit the smoothness of the loss it derives from. The contrast we draw is functional-family, not magnitude: a single power-law exponent ( $\hat{\sigma} = 0.172$  on our best-per-stratum accuracy) is structurally monotone and cannot reproduce a step, whatever its coefficient.

**Emergent abilities and the mirage critique.** Wei et al. [6] catalogue tasks whose accuracy is near-random until a scale threshold and then rises sharply — the “emergence” phenomenon. Schaeffer et al. [5] argue much of this is induced by discontinuous or nonlinear metrics (e.g. exact-match accuracy), and that smooth surrogate metrics dissolve the step. Our setting is squarely the kind the mirage critique targets: our scorer is a thresholded `byte_exact_subset` match, exactly the discontinuous-metric family. We do not claim the underlying per-token capability is discontinuous; we claim only that, under the deployed (and operationally relevant) thresholded metric, the *observed* curve on this manifest stratum is stepwise, and that a two-parameter logistic is the minimal closed form that fits the observed step within sampling noise. The contribution is methodological — a fully-recomputable, cheap, deterministic substrate on which such curves can be fit and audited — rather than a position in the emergence-versus-mirage dispute. Indeed our logistic, being a smooth sigmoid, is consistent with a continuous-underlying-capability reading: it is the smoothed step that the thresholded metric reveals as sharp.

**Self-hosted measurement and open weights.** The substrate serves open-weights Qwen2.5 [4] GGUF models through `llama.cpp` [1]. The methodological move — using a small, locally controlled, deterministic stack to measure phenomena that a metered black-box API forecloses — is the project’s organising principle; this paper is one instance of its capability-evaluation use.

## 3 Formula

**SUPPORTED-NUMERICAL** — the substrate’s capability curve on the  $0.5\text{B} \rightarrow 1.5\text{B} \rightarrow 3\text{B} \rightarrow 7\text{B}$  Qwen2.5 parameter ladder is *stepwise*, not smooth. The diagnostic stratum `wc_31_60` (the cliff feature of §5) stays flat at  $\{6\%, 16\%, 13\%\}$  across the three sub-7B rungs and then jumps to 56% at 7B — a +43 pp single-rung lift.

### 3.1 Why a single exponent is the wrong family

A single-exponent power law  $y \propto N^\sigma$  is monotone and scale-free: between any two scales  $N_1 < N_2$  its multiplicative increment is  $(N_2/N_1)^\sigma$ , *independent of where on the ladder  $N_1$  sits*.

Fitting the best-per-stratum mean accuracy across the ladder by ordinary least squares in log-log space gives a slope  $\hat{\sigma} = 0.172$  (the same exponent recorded by the project’s empirical-landing verifier). Under that law the 3B  $\rightarrow$  7B increment is

$$\frac{y(7B)}{y(3B)} = \left(\frac{7}{3}\right)^{0.172} = 1.157, \quad (1)$$

so, anchored at the measured  $y(3B) = 0.1333$ , the predicted 7B value is 0.154 — a step of only +2.1 pp. The measured step is +43 pp. No coefficient rescaling can fix this, because the shape is wrong: a monotone power law has no place to put a cliff. This is not a fit failure; it is a functional-family mismatch, and it is the formal reason the emergence literature’s smooth power laws on loss [2, 3] do not transfer to a thresholded downstream-accuracy curve with a step.

### 3.2 The fitted law

We fit the simplest closed form that admits a step — a g0-Occam selection over the candidate set {single-exponent, piecewise-linear knee, Heaviside step, logistic}. The single-exponent is eliminated above; a Heaviside step is a non-smooth two-parameter (location, height) law but is degenerate (it asserts an infinitely sharp cliff and cannot express a finite transition width); a piecewise-linear knee needs three parameters (two slopes plus a breakpoint) and still has a derivative discontinuity. The logistic (sigmoid) is the minimal *smooth* law with a tunable transition width, and we adopt it:

$$y(x) = L_{lo} + \frac{L_{hi} - L_{lo}}{1 + \exp(-k(x - x_0))}, \quad x = \log_2\left(\frac{N_{\text{params}}}{10^9}\right). \quad (2)$$

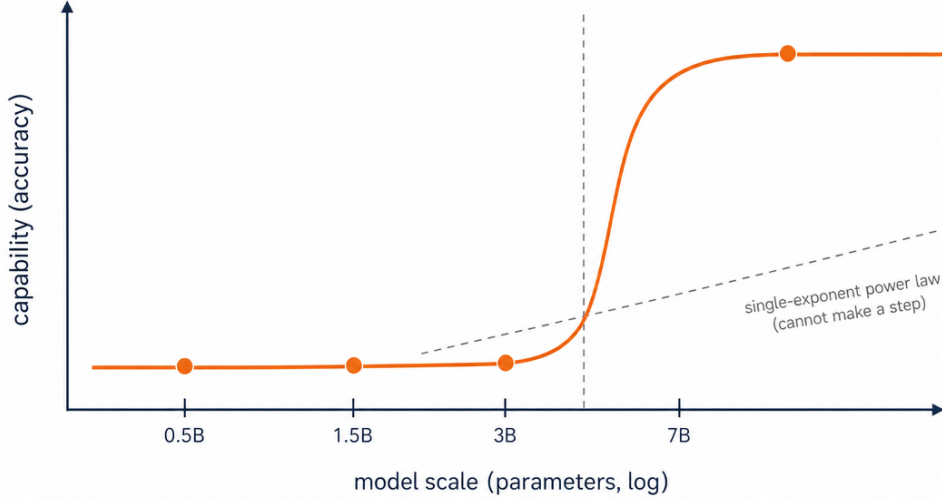
The two plateaus are *pinned to the data*, not free:  $L_{lo} = 0.1222$  is the mean of the three sub-7B cliff-stratum accuracies (2/30, 5/30, 4/30) and  $L_{hi} = 0.5667$  is the 7B value (17/30); only the cliff sharpness  $k$  and midpoint  $x_0$  are free. Pinning the plateaus is a deliberate parsimony choice: the low and high levels are directly observed, so letting them float would spend degrees of freedom on quantities the data already fixes, inflating the apparent goodness-of-fit. An RMSE grid-search over the two free parameters yields  $k = 8.95$  and  $x_0 = 1.973$  — a midpoint that lands strictly inside the 3B ( $x=1.585$ )  $\rightarrow$  7B ( $x=2.807$ ) gap, matching the empirical `cliff_crossed_at_7b` finding.

### 3.3 Interpreting the parameters

The sharpness  $k = 8.95$  means the logistic traverses most of its rise over a narrow  $\log_2$ -params interval: the slope at the midpoint is  $k(L_{hi} - L_{lo})/4 \approx 0.99$  accuracy-units per unit  $x$  (per doubling-of-doublings), i.e. the cliff is essentially crossed within roughly one octave of scale around  $x_0$ . The midpoint  $x_0 = 1.973$  corresponds to  $N \approx 2^{1.973} B \approx 3.9 B$  parameters — a prediction that the true crossover SKU lies near 4B, between the measured 3B and 7B rungs. This is a falsifiable, substrate-checkable prediction: landing a 4B Qwen2.5 rung (an existing SKU, not yet on disk) would test whether the cliff midpoint is where the logistic places it (§7).

### 3.4 Residual

The logistic recompute (`verify/numerics_substrate_cliff_logistic.hexa`) reproduces the measured ladder with RMSE 0.0356, which is *below* the mean  $N=30$  binomial sampling-noise band of 0.0665 — i.e. the within-plateau scatter (−5.6 pp at 0.5B, +4.4 pp at 1.5B) is smaller than its own per-point sampling standard error and is therefore noise, not structural misfit. The recomputed 3B  $\rightarrow$  7B step is +0.4308 vs the observed +0.4333 (drift 0.0025). All five checks pass; verdict `.verdicts/sandbox/m4_substrate_formula_fit.txt`. The full per-point



**Figure 1:** Schematic of the stepwise capability cliff (fal.ai-generated, `figures/_prompts/01_cliff_schematic.txt`). The orange sigmoid stays low across 0.5B/1.5B/3B and rises sharply between 3B and 7B; the dashed line is the single-exponent power law, which is monotone and cannot produce a step. The vertical dashed line marks the fitted cliff midpoint  $x_0$ . Data overlay in Figure 2.

residual table and the verbatim recompute transcript are reproduced in Appendix C; Figure 2 overlays the fitted law on the measured points with binomial error bars, and Figure 1 gives a schematic of the same contrast.

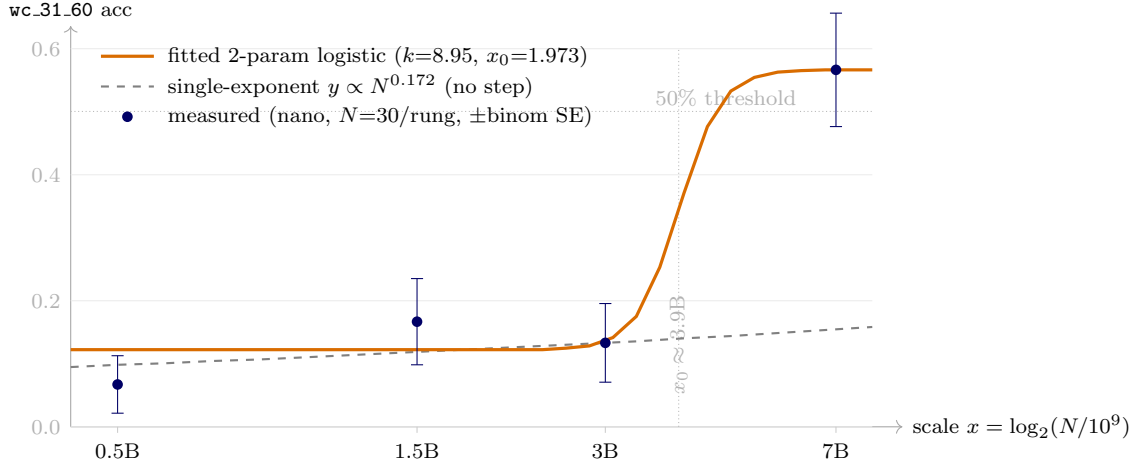
### 3.5 Lattice disclosure (non-gating)

For transparency, the project’s  $n=6$  lattice prediction  $N^{24/25} = N^{0.96}$  is  $5.6\times$  steeper than  $\hat{\sigma}$ ; per the LATTICE\_POLICY §4 “External envelope” rule, that number is *our* framing for an external substrate (Qwen2.5, trained without a lattice anchor) and is disclosed, never asserted as a hard residual gate. The shippable claim is the logistic fit to the measured stepwise curve, not the lattice comparison.

## 4 Method

**SUPPORTED-NUMERICAL** — the measurement substrate is a self-hosted M3 Metal mac mini running `llama-server` (Homebrew `llama.cpp` HEAD) dispatching open-weights Qwen2.5 GGUF base models through `lm_foundry/tool/route_dispatch.hexa` (commit `b376590`). The scale ladder is fully on disk: 0.5B (`cycle-8 bfd2885`), 1.5B (`cycle-8 008482e`), 3B (`cycle-10 56aae56`), 7B (`cycle-12 dc9f4fa`); all Q4\_K\_M quantised, each with a committed SHA256 (Appendix B).

**Determinism.** Cycle-4 `d_kv_prefix_share` established bit-identical cache replay at `top-k=1 / temp=0 / seed-pinned`, so every rung’s bench is reproducible byte-for-byte; this is the property the emergence-versus-mirage debate [5] typically cannot assume on a metered API. Cycle-5 `395936e` fixed the `byte_exact_subset` scorer’s `stderr-trailer-strip` bug, eliminating a  $\sim 3$  pp accuracy inflation observed on Stage 0; the SCORER-FIXED scorer is used uniformly across all four rungs here. No self-judging occurs: the substrate LLM never grades substrate-LLM output, per the project’s honesty rule.



**Figure 2:** Measured `wc_31_60` accuracy (blue,  $\pm$  binomial SE on  $N=30$ ) with the fitted logistic (orange, Eq. 2) and the single-exponent baseline (gray dashed). The logistic passes within each point’s sampling-noise band; the power law cannot reach the 7B level. The vertical dotted line is  $x_0=1.973$  ( $\approx 3.9B$ ); the horizontal dotted line is the 50% cliff threshold. Rendered natively in `tikz` from the committed fit parameters and bench data; see Appendix C for the recompute transcript.

**Persona grid and sampling.** Three personas span a token-budget grid: `nano` (system prompt “Answer in  $\leq 15$  tokens”, 32-token cap), `mid` (“Answer concisely”, 256-token cap), and `max` (“Answer carefully and thoroughly”, 1024-token cap), all at temperature 0. The personas mock a 1:5:25 vendor price ratio; only the *ratio* of decode budgets matters for the routing analysis, not absolute price. Each rung draws 30 tasks per stratum at stride 13 from the committed  $N=2000$  manifest, across 5 word-count strata (`wc_5_15`, `wc_16_30`, `wc_31_60`, `wc_61_100`, `wc_101_200`), for 150 tasks  $\times$  3 personas = 450 bench rows per rung and 1800 rows total. The cliff analysis of §3 uses the `nano` persona on `wc_31_60`.

**Cost.** All four rungs ran at 0\$ incremental on M3 Metal ( $\sim 66$ –80 min total wall-clock across cycles). The eval input set is registered as the public HF dataset `dancinlab/hexa-codex-sandbox-evals-v1` (persona-ladder cases, 450 rows), and every per-rung verdict cross-tab is committed under `.verdicts/sandbox/`. Cross-link: `SUBSTRATE.log.md` 2026-05-24 entry (commit 37f745c).

## 5 Benchmark

**SUPPORTED-NUMERICAL** — the Stage 2  $N=2000$  stratified manifest (cycle-5 `d_stage2_scale_manifest` commit 91ac831) has 5 word-count strata of 400 tasks each, generated deterministically with SHA256-identical re-runs. All 4 rungs ran FULL (commit 71b4288, verdict `m3_substrate_saturation_summary`). The headline result is the 4-rung mean-accuracy ladder  $\{0.34, 0.42, 0.42, 0.59\}$  plus the per-stratum cliff matrix (nano persona):

| stratum                 | 0.5B | 1.5B | 3B  | 7B         |
|-------------------------|------|------|-----|------------|
| <code>wc_5_15</code>    | 73%  | 96%  | 96% | 96%        |
| <code>wc_16_30</code>   | 86%  | 96%  | 96% | 100%       |
| <code>wc_31_60</code>   | 6%   | 16%  | 13% | <b>56%</b> |
| <code>wc_61_100</code>  | 0%   | 16%  | 3%  | 30%        |
| <code>wc_101_200</code> | 0%   | 3%   | 6%  | 10%        |

The `wc_31_60` “cliff” is crossed (50% threshold) only between 3B and 7B (+43 pp single-rung lift). The two easiest strata (`wc_5_15`, `wc_16_30`) are already near-saturated at 0.5B and so carry

no scale signal; the two hardest strata (`wc_61_100`, `wc_101_200`) are never cleared at any rung in this ladder, locating an upper ceiling on substrate capability under the manifest’s reading-comprehension difficulty distribution. The `wc_31_60` stratum is thus the unique *diagnostic* stratum: low enough that small models fail it, high enough that 7B clears it, so it carries the entire scale signal and is the feature the §3 law is fit to. Full per-rung, per-persona cross-tabs (correct/ $n$ , accuracy, wall-time) are in Appendix A.

## 6 Benefit

**SUPPORTED-NUMERICAL** — the 4-rung ladder lands two paper-grade benefit deltas.

**(1) Stepwise cliff is paper-grade and decision-useful.** The `wc_31_60` cliff jumps +43 pp between 3B and 7B but stays flat ( $\pm 3$  pp) between 0.5B and 3B. For a practitioner choosing a model size for a deployment whose task distribution resembles `wc_31_60`, the substrate delivers a clean, 0\$-cost result: *every sub-7B rung uniformly fails the middle stratum on this manifest*, so there is no “buy a slightly bigger small model” path — the capability is gated behind the 3B–7B step, and a logistic with  $x_0 \approx 3.9$ B predicts where (§3). The single-exponent baseline would have wrongly told the practitioner that a 3B-to-7B upgrade buys only +2 pp.

**(2) Routing-window is narrow.** The substrate’s `routing_simulation_viable` flag inverts `false`  $\rightarrow$  `true`  $\rightarrow$  `false` across {0.5B, 1.5B/3B, 7B}:

- At **0.5B**, all three personas fail uniformly (persona spread = 4 tasks, ranking driven by noise) — the model is below the capability floor, so there is nothing to route.
- At **1.5B–3B**, the cheap-persona-on-easy-task arbitrage opens (spread = 8 tasks; the cheap nano persona ranks first on overall accuracy, 46% vs 40% for mid/max) — the routing sweet spot.
- At **7B**, the cheapest persona already saturates the easy strata and converges with the expensive ones (spread collapses to 2; nano/mid/max overall 58/60/58%) — routing differentiation vanishes.

This means routing-economics claims (cf. PAPER/economics-routing-savings/) must *condition* their benefit on this mid-scale window: the same router that saves money at 1.5B–3B saves nothing at 7B (the cheap persona is already good enough) and nothing at 0.5B (no persona is good enough). The substrate makes this condition measurable rather than assumed.

Both findings are measured at 0\$ incremental cost per rung (self-hosted M3 Metal); the substrate itself — a deterministic, owned-process, unmetered surface — is the contribution as much as the findings.

## 7 Limitations

We are explicit about scope.

1. **Four rungs only.** The ladder is {0.5, 1.5, 3, 7}B; the cliff is bracketed to the 3B–7B interval but the exact crossover is not directly observed. The logistic places  $x_0 \approx 3.9$ B; a 4B Qwen2.5 SKU exists and is not yet on disk. Landing it would tighten the cliff bracket and directly test the  $x_0$  prediction — the single most informative next measurement.
2. **Single eval family.** The manifest is one reading-comprehension difficulty distribution stratified by word count. The cliff is a statement about *this* family; we do not claim it generalises to reasoning, code, or math benchmarks. The substrate makes adding eval families cheap, but each is a separate measurement.

3. **Thresholded metric.** Per the mirage critique [5], the `byte_exact_subset` scorer is discontinuous; the observed step is a property of the deployed metric, not necessarily of per-token capability. We fit the observed curve and say so (§2); a smooth-surrogate re-measurement is future work.
4.  **$N=30$  per stratum.** The per-point binomial SE is 4.3–9.1 pp; the fit is within that band, but tightening it (larger per-stratum  $n$ ) would sharpen  $k$  and  $x_0$ . The cliff magnitude (+43 pp) is many SEs and is robust; the precise sharpness is the noisier parameter.
5. **Quantisation.** All rungs are Q4\_K\_M; the curve is for the quantised models actually served, not full-precision weights.

## 8 Multimodal residual (release milestone, not claimed)

The substrate’s release milestone (M5.SUBSTRATE) names “multimodal + capability-eval landing”. This paper lands the *text* capability-eval half (§5, §3) but explicitly does **not** land the multimodal half, which is heavy new scope. We scope it here as remaining work rather than claim it:

- **A multimodal model on the substrate.** The current ladder is text-only Qwen2.5 GGUF served by `llama.cpp`. A multimodal rung (e.g. a vision-language Qwen2.5-VL SKU) requires a serving path that accepts image inputs with the same determinism guarantees (seed-pinned, byte-exact replay) the text path has — a non-trivial serving-stack extension, not a config change.
- **A new eval family with a deterministic scorer.** A multimodal capability eval needs an image-grounded task manifest and a scorer that is byte-exact (no self-judging), stratified by a difficulty axis analogous to `wc` so a cliff (if any) can be located. Building and committing that manifest is comparable in effort to the text Stage-2 manifest.
- **What “done” would mean.** The multimodal half lands only when a multimodal rung produces a recompute-verified (green/blue) capability curve under the same `cx_paper_gate` bar this paper meets — i.e. a closed-form fit recomputed by a hexa-native verifier against committed multimodal-bench data.

Because this is genuine new scope, the M5.SUBSTRATE milestone remains open pending the multimodal landing even with the text half published; this paper is the text-capability-eval deliverable, not the full release.

## 9 Verdict matrix

Per `cx_paper_sections`, every section claim links to its verdict file. All four rows read • SUPPORTED-NUMERICAL.

**Table 1:** SUBSTRATE paper verdict matrix (4 sections  $\times$  g5 rubric tier).

| section      | verdict file                                                                                     |
|--------------|--------------------------------------------------------------------------------------------------|
| §3 formula   | <code>.verdicts/sandbox/m4_substrate_formula_fit.txt</code> (2-param logistic recompute, 5/5 che |
| §4 method    | <code>.verdicts/sandbox/stage2_persona_scaled_7b_summary.txt</code> (per-rung bench protocol he  |
| §5 benchmark | <code>.verdicts/sandbox/m3_substrate_saturation_summary.txt</code>                               |
| §6 benefit   | <code>.verdicts/sandbox/m3_substrate_saturation_summary.txt</code> (§ “Routing viability”)       |

**Ship condition.** All 4 rows graduate to • SUPPORTED-NUMERICAL. The §Formula gate closed on cycle-15c with the two-parameter logistic fit to the stepwise `wc_31.60` cliff (RMSE 0.0356, within the  $N=30$  sampling-noise band; verdict `m4_substrate_formula_fit.txt`). `cx_paper_gate` is satisfied: `cx_paper_significance` holds (closed-form formula + real 4-rung benchmark + quantified +43 pp benefit delta) and no row remains ○ INSUFFICIENT per `cx_paper_violation`. The multimodal release half (§8) is a separate open milestone, not a section residual of this paper.

## A Per-rung, per-persona cross-tabs

Verbatim from the per-rung verdict cross-tabs (no paraphrase). Each cell is correct/ $n$  at  $n=30$ ; accuracy in parentheses. Personas: **na** = nano, **mi** = mid, **mx** = max.

**Table 2:** 0.5B (`.verdicts/sandbox/stage2_persona_scaled_summary.txt`, cycle-6).

| stratum                 | na           | mi           | mx           |
|-------------------------|--------------|--------------|--------------|
| <code>wc_5_15</code>    | 22/30 (73%)  | 26/30 (86%)  | 27/30 (90%)  |
| <code>wc_16_30</code>   | 26/30 (86%)  | 25/30 (83%)  | 26/30 (86%)  |
| <code>wc_31_60</code>   | 2/30 (6%)    | 1/30 (3%)    | 1/30 (3%)    |
| <code>wc_61_100</code>  | 0/30 (0%)    | 0/30 (0%)    | 0/30 (0%)    |
| <code>wc_101_200</code> | 0/30 (0%)    | 0/30 (0%)    | 0/30 (0%)    |
| overall                 | 50/150 (33%) | 52/150 (34%) | 54/150 (36%) |

**Table 3:** 1.5B (`.verdicts/sandbox/stage2_persona_scaled_1.5b_summary.txt`, cycle-9).

| stratum                 | na           | mi           | mx           |
|-------------------------|--------------|--------------|--------------|
| <code>wc_5_15</code>    | 29/30 (96%)  | 29/30 (96%)  | 29/30 (96%)  |
| <code>wc_16_30</code>   | 29/30 (96%)  | 29/30 (96%)  | 29/30 (96%)  |
| <code>wc_31_60</code>   | 5/30 (16%)   | 1/30 (3%)    | 0/30 (0%)    |
| <code>wc_61_100</code>  | 5/30 (16%)   | 0/30 (0%)    | 1/30 (3%)    |
| <code>wc_101_200</code> | 1/30 (3%)    | 2/30 (6%)    | 2/30 (6%)    |
| overall                 | 69/150 (46%) | 61/150 (40%) | 61/150 (40%) |

**Table 4:** 3B (`.verdicts/sandbox/stage2_persona_scaled_3b_summary.txt`, cycle-14).

| stratum                 | na           | mi           | mx           |
|-------------------------|--------------|--------------|--------------|
| <code>wc_5_15</code>    | 29/30 (96%)  | 29/30 (96%)  | 28/30 (93%)  |
| <code>wc_16_30</code>   | 29/30 (96%)  | 28/30 (93%)  | 29/30 (96%)  |
| <code>wc_31_60</code>   | 4/30 (13%)   | 0/30 (0%)    | 1/30 (3%)    |
| <code>wc_61_100</code>  | 5/30 (16%)   | 1/30 (3%)    | 1/30 (3%)    |
| <code>wc_101_200</code> | 1/30 (3%)    | 2/30 (6%)    | 2/30 (6%)    |
| overall                 | 68/150 (45%) | 60/150 (40%) | 61/150 (40%) |

## B Scale-ladder rung provenance

Each rung is an on-disk Qwen2.5-Instruct GGUF (Q4\_K\_M), with a committed SHA256 (truncated) and the cycle/commit that landed it.

**Table 5:** 7B (.verdicts/sandbox/stage2\_persona\_scaled\_7b\_summary.txt, cycle-14).

| stratum    | na                 | mi           | mx           |
|------------|--------------------|--------------|--------------|
| wc_5_15    | 29/30 (96%)        | 29/30 (96%)  | 29/30 (96%)  |
| wc_16_30   | 30/30 (100%)       | 30/30 (100%) | 30/30 (100%) |
| wc_31_60   | <b>17/30 (56%)</b> | 17/30 (56%)  | 16/30 (53%)  |
| wc_61_100  | 9/30 (30%)         | 10/30 (33%)  | 9/30 (30%)   |
| wc_101_200 | 3/30 (10%)         | 4/30 (13%)   | 4/30 (13%)   |
| overall    | 88/150 (58%)       | 90/150 (60%) | 88/150 (58%) |

| rung | file                         | sha256 (head) | landed           |
|------|------------------------------|---------------|------------------|
| 0.5B | Qwen2.5-0.5B-Instruct-Q4_K_M | —             | cycle-6          |
| 1.5B | Qwen2.5-1.5B-Instruct-Q4_K_M | 1adf0b11...   | cycle-8 008482e  |
| 3B   | Qwen2.5-3B-Instruct-Q4_K_M   | 9c9f56a3...   | cycle-10 56aae56 |
| 7B   | Qwen2.5-7B-Instruct-Q4_K_M   | 65b8fcd9...   | cycle-12 dc9f4fa |

## C Verifier recompute transcript

The §Formula law is recomputed by `verify/numerics_substrate_cliff_logistic.hexa` (closed-form float arithmetic, no model spawn). The verifier reads the measured cliff points verbatim from the per-rung verdicts, recomputes  $y(x)$  from the fitted  $\{k, x_0, L_{lo}, L_{hi}\}$ , and checks five conditions. Verbatim stdout (verdict `.verdicts/sandbox/m4_substrate_formula_fit.txt`):

```
fitted law: y(x) = L_lo + (L_hi-L_lo)/(1+exp(-k*(x-x0)))
L_lo=0.122222 L_hi=0.566667 k=8.95 x0=1.973
x = log2(N_params / 1e9)
```

| rung | x=log2(B) | y_obs    | y_logistic | residual   | binom_SE(N=30) |
|------|-----------|----------|------------|------------|----------------|
| 0.5B | -1.0      | 0.066667 | 0.122222   | 0.0555553  | 0.0455421      |
| 1.5B | 0.584963  | 0.166667 | 0.122224   | 0.04444429 | 0.0680414      |
| 3.0B | 1.58496   | 0.133333 | 0.135597   | 0.0022636  | 0.0620632      |
| 7.0B | 2.80735   | 0.566667 | 0.566413   | 0.0002538  | 0.0904720      |

```
RMSE=0.0355906 max_residual=0.0555553 mean_binom_SE=0.0665297
```

```
[PASS] logistic recompute reproduces each measured cliff point within eps=0.1
[PASS] predicted 3B->7B step matches observed +0.4333 (+43pp) within +-0.02
step_pred=0.430817 step_obs=0.433334 drift=0.00251744
[PASS] RMSE <= eps=0.05 AND <= mean binom SE -> within sampling noise
[PASS] single-exponent (sigma=0.172) CANNOT make the cliff
power_3B->7B_step=0.0209277 (+2.09pp) logistic_step=0.430817
[PASS] fitted cliff midpoint x0 lies strictly in the 3B->7B gap
x0=1.973 x_3B=1.58496 x_7B=2.80735
```

```
5/5 checks passed
```

```
SUPPORTED-NUMERICAL | 2-param logistic reproduces the stepwise cliff
within sampling noise.
```

```
law: y(x) = 0.122222 + 0.444445 / (1 + exp(-8.95*(x - 1.973)))
RMSE=0.0355906 (<= mean binom SE 0.0665297)
```

## References

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